A Minimal SST Toy Model for Delay-Coupled Correlations on Shared Circulation Carriers: A Reduced Closure Program for the Delay, Coupling, and Coherence Scales

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Abstract

We formulate a minimal Swirl–String Theory (SST) toy model for delay-coupled correlations on shared circulation carriers. The central proposal is that correlated subsystems may be modeled not as independent pointlike entities linked only by an abstract nonlocal state, but as localized readout sites on a common circulation carrier embedded in the swirl medium. At the reduced effective level, the dynamics are described by a delayed phase system with three macroscopic parameters: a transport delay τ, a phase-locking rate κ, and a coherence length ℓ_τ. We show that the canonical SST identification

$$\omega_c = \frac{\|\mathbf{v}_{\circlearrowleft}\|}{r_c}$$

immediately fixes the transport time scale and strongly constrains the remaining two parameters to the form

$$\tau = \frac{L}{\|\mathbf{v}_{\circlearrowleft}\|}, \qquad \kappa_{AB} = \omega_c\,\mathcal{O}_{AB}, \qquad \ell_\tau = r_c\,\mathcal{Q},$$

where $\mathcal{O}_{AB}$ is a dimensionless shared-circulation overlap factor and $\mathcal{Q}$ is a dimensionless phase-quality factor. In the present article, "entanglement-like" refers only to nontrivial angle-dependent correlations between spatially separated readout sites on a common carrier; no claim is made here to a completed derivation of Bell/CHSH phenomenology or the Born rule. We distinguish derived results, closure assumptions, and open problems, propose minimal falsifiers for each sector, and provide a reduced observable construction together with a worked coherence estimate based on topological leakage.

Keywords: Swirl–String Theory, entanglement-like correlations, toy model, delay-coupled phase dynamics, circulation overlap, coherence length, hydrodynamic-topological model

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1 Introduction

The standard quantum formalism represents entanglement through nonfactorizable states in Hilbert space. While empirically successful, that description leaves open the ontological status of the correlation carrier. Hydrodynamic and stochastic reformulations of quantum structure have a substantial history, ranging from Madelung's continuum rewriting of the Schrödinger equation to Nelson's stochastic mechanics, Bohm's hidden-variable program, and modern analogue-gravity or superfluid-vacuum programs [8, 9, 12, 10, 11]. The present article does not attempt to review that literature comprehensively. Instead, it isolates one specific reduced question within SST.

In SST, the primitive ontology is different: the vacuum is modeled as an incompressible, inviscid structured medium, and localized matter-like excitations are topological swirl strings rather than point particles. Within such a framework, it is natural to ask whether nontrivial long-range correlations can be re-expressed, at least in a reduced model, as dynamical consequences of shared circulation structure, finite propagation, and phase-locking.

The present article does not claim a completed derivation of Bell correlations from first principles. Rather, it isolates the minimum dynamical closure problem in the form of a toy model. If correlated subsystems A and B are treated as local phase-bearing sites on a common circulation carrier, then the reduced effective theory is determined by three scales:

(i) a transport delay τ,

(ii) a phase-locking rate κ,

(iii) a coherence length ℓ_τ.

Instead, it isolates a minimal mechanistic sector whose role is to determine whether SST admits a nontrivial reduced model for delay, geometric coupling, and finite coherence on shared circulation carriers.

Throughout this article, the phrase "entanglement-like" is used only in this reduced operational sense: separated readout sites on a shared carrier may display angle-dependent correlations not attributable to independent local phases, without thereby claiming a completed reproduction of quantum-mechanical entanglement.

The main result of this paper is that SST already fixes the parametric structure of all three at the level of this toy model:

$$\tau = \frac{L}{\|\mathbf{v}_{\circlearrowleft}\|}, \qquad \kappa_{AB} = \omega_c\,\mathcal{O}_{AB}, \qquad \ell_\tau = r_c\,\mathcal{Q}.$$

What is specific to SST in this reduction is not merely the presence of a characteristic frequency, but the fact that the transport, coupling, and coherence sectors are all tied back to the same primitive core scales $(r_c, \|\mathbf{v}_{\circlearrowleft}\|)$ through the identifications $\omega_c = \|\mathbf{v}_{\circlearrowleft}\|/r_c$, $\kappa_{AB} = \omega_c\mathcal{O}_{AB}$, and $\ell_\tau = r_c\mathcal{Q}$.

The only remaining physical inputs are geometric overlap and the mechanism of phase-memory loss.

The purpose of the paper is therefore not to replace the quantum formalism, but to ask whether a minimal hydrodynamic-topological model can isolate a plausible SST mechanism for transport delay, geometric coupling, and finite coherence on shared circulation carriers.

2 Epistemic status of the claims

To keep the logical status explicit, we separate the statements as follows.

- **Derived from current SST anchors:**

$$\omega_c = \frac{\|\mathbf{v}_{\circlearrowleft}\|}{r_c}, \qquad \tau = \frac{L}{\|\mathbf{v}_{\circlearrowleft}\|}.$$

- **Constrained toy-model closure forms:**

$$\kappa_{AB} = \omega_c \, \mathcal{O}_{AB}, \qquad \ell_\tau = r_c \, \mathcal{Q}.$$

- **Modeled but not yet first-principles derived:** explicit formulas for $\mathcal{O}_{AB}$ and $\mathcal{Q}$.
- **Open program:** recovery of exact Bell/CHSH statistics and operational no-signaling from the underlying SST medium dynamics.
- **Not claimed here:** equivalence to standard quantum mechanics, a derivation of the Born rule, or a full SST account of quantum measurement.

3 Canonical SST anchors

For the purposes of the present toy model, SST is taken to supply three working primitives: a core radius r_c, a characteristic swirl speed $\|\mathbf{v}_{\circlearrowleft}\|$, and a fundamental circulation scale. These are adopted here as model inputs rather than derived within the present article. In the broader SST program, these quantities are calibrated to the electron/Compton/core closure chain; in the present paper they function only as reduced-model constants.

Their intended provenance may be summarized briefly as follows. SST adopts the classical electron radius

$$r_e = \frac{\alpha \hbar}{m_e c}, \tag{1}$$

and takes the core radius to be

$$r_c = \frac{r_e}{2}. \tag{2}$$

The characteristic angular rate is identified with the electron Compton angular frequency

$$\omega_c = \frac{m_e c^2}{\hbar}, \tag{3}$$

and the characteristic swirl speed is then fixed by the kinematic closure

$$\|\mathbf{v}_{\circlearrowleft}\| = \omega_c r_c. \tag{4}$$

The present article does not defend that broader closure program; it uses only the resulting reduced constants.

The fundamental circulation is

$$\Gamma_0 = 2\pi r_c \|\mathbf{v}_{\circlearrowleft}\|. \tag{5}$$

The canonical Compton-frequency identification is

$$\omega_c = \frac{\|\mathbf{v}_{\circlearrowleft}\|}{r_c}. \tag{6}$$

which is dimensionally exact:

$$[\omega_c] = \frac{\mathrm{m\,s^{-1}}}{\mathrm{m}} = \mathrm{s^{-1}}.$$

Using the canonical constants

$$\|\mathbf{v}_{\circlearrowleft}\| = 1.09384563 \times 10^6 \ \mathrm{m\,s^{-1}}, \qquad r_c = 1.40897017 \times 10^{-15} \ \mathrm{m},$$

one obtains

$$\omega_c = \frac{1.09384563 \times 10^6}{1.40897017 \times 10^{-15}} \approx 7.76344 \times 10^{20} \ \mathrm{s^{-1}}. \tag{7}$$

This identification is crucial because it fixes the intrinsic turnover time of an elementary circulation core within the toy model:

$$\omega_c^{-1} \approx 1.288 \times 10^{-21} \ \mathrm{s}. \tag{8}$$

4 Effective delayed phase model

We model two correlated sites A and B by phase variables $\phi_A(t)$ and $\phi_B(t)$ satisfying the symmetric delayed phase system

$$\dot{\phi}_A(t) = \omega_0 + \kappa_{AB} \sin[\phi_B(t-\tau) - \phi_A(t)], \tag{9}$$

$$\dot{\phi}_B(t) = \omega_0 + \kappa_{AB} \sin[\phi_A(t-\tau) - \phi_B(t)]. \tag{10}$$

The structure of equations (9) and (10) is intentionally orthodox: it is a finite-delay phase-locking model. The SST content lies not in the existence of such equations, but in the interpretation of τ, κ, and ℓ_τ as hydrodynamic-topological quantities.

Define the phase difference

$$\Delta(t) = \phi_A(t) - \phi_B(t). \tag{11}$$

Subtracting equations (9) and (10) yields

$$\dot{\Delta}(t) = -2\kappa_{AB} \sin\big(\Delta(t-\tau)\big). \tag{12}$$

Stationary branches satisfy

$$\sin(\Delta^*) = 0 \quad \Rightarrow \quad \Delta^* = n\pi, \qquad n \in \mathbb{Z}. \tag{13}$$

Thus a delay-coupled closed phase system admits discrete branch selection even before any quantum-statistical interpretation is imposed.

5 Reduced observables and correlation function

To make the toy model operational, we define binary readout observables by thresholding the local phases against analyzer angles θ_A and θ_B:

$$A(\theta_A, t) = \mathrm{sign}\big[\cos(\phi_A(t) - \theta_A)\big], \tag{14}$$

$$B(\theta_B, t) = \mathrm{sign}\big[\cos(\phi_B(t) - \theta_B)\big]. \tag{15}$$

These observables take values in $\{-1, +1\}$ and define the reduced-model correlation function

$$C(\theta_A, \theta_B) := \langle A(\theta_A, t)\, B(\theta_B, t)\rangle_t\,, \tag{16}$$

where $\langle\cdot\rangle_t$ denotes a long-time average on a stationary branch or an ensemble average over initial conditions within the toy model.

Equations equations (14) to (16) do not constitute a Bell/CHSH framework by themselves. Their purpose is narrower: they provide an explicit observable layer showing how the delayed phase system can generate nontrivial angle-dependent correlations in a self-contained reduced setting.

6 Derivation of the transport delay

Consider a phase disturbance propagating along a circulation structure of arclength L. If the transport speed is the characteristic SST swirl speed $\|\mathbf{v}_\circlearrowleft\|$, then the delay is simply

$$\boxed{\tau = \frac{L}{\|\mathbf{v}_\circlearrowleft\|}.} \tag{17}$$

This is the natural kinematic first-pass closure consistent with the canonical identification $\omega_c = \|\mathbf{v}_\circlearrowleft\|/r_c$.

Dimensional check:

$$[\tau] = \frac{\mathrm{m}}{\mathrm{m\,s^{-1}}} = \mathrm{s}.$$

Equivalently,

$$\tau\,\omega_c = \frac{L}{\|\mathbf{v}_{\circlearrowleft}\|}\frac{\|\mathbf{v}_{\circlearrowleft}\|}{r_c} = \frac{L}{r_c}. \tag{18}$$

Thus the delay measured in carrier cycles is simply the geometric loop length in units of the core radius.

Interpretation

Equation (18) is important. It says that a circulation loop of length L contains L/r_c core-turnover lengths, so the delay is not an independent free time parameter but a geometric bookkeeping of how many core scales fit around the carrier.

7 Why the coupling rate must be geometric

In equations (9) and (10), κ multiplies a sine and must therefore have units of inverse time:

$$[\kappa] = \mathrm{s}^{-1}.$$

Since ω_c is the only intrinsic SST primitive with those units, κ must take the form

$$\kappa = \omega_c \times (\text{dimensionless factor}). \tag{19}$$

That dimensionless factor cannot be fixed by ρ_f alone. Density sets energetic scale, but not the degree to which two localized phase probes sample the same circulation object. Therefore κ must depend on geometry. In the present SST reading, this geometric dependence should not be interpreted as mere spatial proximity. Because the overlap is built from the dot product $\mathbf{u}_A \cdot \mathbf{u}_B$, it encodes both shared support in space and directional alignment of the carrier fields. The reduced coupling factor $\mathcal{O}_{AB}$ is therefore intended to capture carrier compatibility in both position and orientation, rather than only naive co-location.

We introduce the *shared-circulation overlap factor*

$$\mathcal{O}_{AB} \in [-1, 1], \tag{20}$$

and write the general SST closure

$$\boxed{\kappa_{AB} = \omega_c\,\mathcal{O}_{AB}.} \tag{21}$$

8 Geometric derivation of the overlap factor

The heuristic closure

$$\mathcal{O}_{AB} \approx \frac{\ell_{\text{shared}}}{L}$$

can be strengthened by deriving the overlap factor directly from the SST interaction energy of two circulation carriers.

Let $\mathbf{u}_A(\mathbf{x})$ and $\mathbf{u}_B(\mathbf{x})$ denote the velocity fields induced by two localized phase-carrying swirl structures A and B. The natural SST interaction energy is

$$E_{AB}^{\text{int}} = \rho_f \int_{\mathbb{R}^3} \mathbf{u}_A(\mathbf{x}) \cdot \mathbf{u}_B(\mathbf{x})\, d^3x, \tag{22}$$

with self-energies

$$E_A = \rho_f \int_{\mathbb{R}^3} \|\mathbf{u}_A(\mathbf{x})\|^2\, d^3x, \qquad E_B = \rho_f \int_{\mathbb{R}^3} \|\mathbf{u}_B(\mathbf{x})\|^2\, d^3x. \tag{23}$$

This suggests the normalized overlap functional

$$\boxed{\mathcal{O}_{AB} := \frac{\displaystyle\int_{\mathbb{R}^3} \mathbf{u}_A \cdot \mathbf{u}_B \, d^3x}{\sqrt{\left(\displaystyle\int_{\mathbb{R}^3} \|\mathbf{u}_A\|^2 \, d^3x\right)\left(\displaystyle\int_{\mathbb{R}^3} \|\mathbf{u}_B\|^2 \, d^3x\right)}}} \tag{24}$$

which is dimensionless and bounded by

$$-1 \le \mathcal{O}_{AB} \le 1 \tag{25}$$

by the Cauchy–Schwarz inequality.

Transparency of the reduction chain. The approximation hierarchy used in this section is as follows. At the canonical level, $\mathcal{O}_{AB}$ is defined by the normalized cross-energy overlap equation (24). The reduction to equation (31) assumes thin carriers with identical transverse profiles and aligned tangent directions on the shared segment. The final form ℓ_{shared}/L is the equal-length specialization $L_A = L_B$. If these assumptions fail, the canonical overlap definition remains valid, but the simplified shared-length reduction no longer need hold.

It is important that this hierarchy be read as a progression from a field-based definition to increasingly specialized geometric limits. The simplest shared-length picture is therefore not the primary SST object, but the most compressed reduction of a more general carrier-compatibility functional. In particular, once tangent alignment, chirality, or local phase transport differ appreciably between two sectors, the overlap must be understood as a genuine field overlap rather than a pure arclength fraction.

The coupling rate is then

$$\boxed{\kappa_{AB} = \omega_c \, \mathcal{O}_{AB}.} \tag{26}$$

8.1 Thin-filament reduction

To connect equation (24) with the simpler shared-length closure, assume both carriers are thin filaments with the same transverse core profile $f(\mathbf{x}_\perp)$, equal characteristic amplitude u_0, and aligned tangent direction $\hat{\mathbf{t}}(s)$ on the shared segment:

$$\mathbf{u}_A(\mathbf{x}) \approx u_0 \, f(\mathbf{x}_\perp) \, \chi_A(s) \, \hat{\mathbf{t}}(s), \qquad \mathbf{u}_B(\mathbf{x}) \approx u_0 \, f(\mathbf{x}_\perp) \, \chi_B(s) \, \hat{\mathbf{t}}(s), \tag{27}$$

where χ_A, χ_B are support functions along arc length s.

Then

$$\int \mathbf{u}_A \cdot \mathbf{u}_B \, d^3x = u_0^2 \left(\int |f|^2 \, d^2x_\perp\right)\left(\int \chi_A(s)\chi_B(s) \, ds\right), \tag{28}$$

while

$$\int \|\mathbf{u}_A\|^2 \, d^3x = u_0^2 \left(\int |f|^2 \, d^2x_\perp\right) L_A, \tag{29}$$

$$\int \|\mathbf{u}_B\|^2 \, d^3x = u_0^2 \left(\int |f|^2 \, d^2x_\perp\right) L_B. \tag{30}$$

Therefore

$$\boxed{\mathcal{O}_{AB} = \frac{\ell_{\text{shared}}}{\sqrt{L_A L_B}},} \tag{31}$$

where

$$\ell_{\text{shared}} = \int \chi_A(s)\chi_B(s) \, ds.$$

For equal loop lengths $L_A = L_B = L$, equation (31) reduces to

$$\boxed{\mathcal{O}_{AB} = \frac{\ell_{\mathrm{shared}}}{L}.} \tag{32}$$

Thus the earlier shared-length law is not a separate ansatz but the equal-length thin-filament limit of the energy-normalized SST overlap functional.

8.2 Sign and chirality

Because $\mathcal{O}_{AB}$ is built from $\mathbf{u}_A \cdot \mathbf{u}_B$, it is sensitive to orientation:

$$\mathcal{O}_{AB} > 0 \quad \text{for co-oriented circulation,}$$
$$\mathcal{O}_{AB} < 0 \quad \text{for anti-oriented circulation,}$$
$$\mathcal{O}_{AB} \approx 0 \quad \text{for orthogonal or disjoint carriers.}$$

Accordingly, the effective phase equation may be written in the signed form

$$\dot{\phi}_A = \omega_0 + \omega_c\, \mathcal{O}_{AB} \sin[\phi_B(t-\tau) - \phi_A(t)], \tag{33}$$

with an analogous equation for ϕ_B. This retains the chirality-sensitive structure of the overlap rather than absorbing it into a positive scalar coupling.

8.3 Physical reading

Equation (26) implies that correlation strength is not primitive. In the toy model, it is controlled by the normalized overlap of genuinely shared circulation structure. In SST language, two subsystems are strongly correlated not because they are assigned a mysterious abstract relation, but because they are modeled as two local manifestations of the same phase-carrying object.

This also suggests a refinement of interpretation: $\mathcal{O}_{AB}$ should be read not only as a measure of shared support in space, but more generally as a reduced measure of geometric and kinematic compatibility on a common carrier. At minimum this includes shared support, tangent alignment, and orientation; in more refined versions it may also include resonant compatibility of the local phase mode. The present article does not yet build such resonance weighting explicitly, but the overlap formalism is intended to leave room for that stronger interpretation.

8.4 A minimal explicit example

The overlap functional $\mathcal{O}_{AB}$ becomes concrete once explicit carrier fields are chosen. As the simplest reduced example, consider two identical coaxial thin carriers with the same transverse profile $f(\mathbf{x}_\perp)$, the same characteristic amplitude u_0, equal arclength L, and fully overlapping support along the carrier coordinate s. Then $\chi_A(s) = \chi_B(s)$, so $\ell_{\mathrm{shared}} = L$, and equation (32) gives

$$\mathcal{O}_{AB} = 1 \tag{34}$$

for co-oriented circulation.

If the same two carriers are anti-oriented, the sign of $\mathbf{u}_A \cdot \mathbf{u}_B$ reverses everywhere in the overlap region and one obtains

$$\mathcal{O}_{AB} = -1. \tag{35}$$

In the opposite limit of disjoint or ideally orthogonal carriers, the overlap becomes small and

$$\mathcal{O}_{AB} \approx 0. \tag{36}$$

A slightly less degenerate intermediate case is two equal-length coaxial carriers with only partial shared support. If the common support occupies a fraction $f \in (0,1)$ of the total arclength, so that

$$\ell_{\text{shared}} = fL,$$

then equation (32) gives

$$\mathcal{O}_{AB} = f. \tag{37}$$

These examples are intentionally minimal. They do not yet constitute a full Biot–Savart evaluation for realistic SST knot geometries. Their role is narrower: they show how the abstract overlap functional maps onto the simplest carrier configurations and how its sign tracks co- versus anti-oriented circulation.

9 Why the coherence length needs a physical decay channel

Here and below, "entanglement-like" continues to mean only the reduced operational notion introduced in the abstract and Introduction: nontrivial angle-dependent correlations between separated readout sites on a shared carrier.

Suppose the two-point correlation takes the phenomenological form

$$E(\theta_A, \theta_B) \sim e^{-L/\ell_\tau} \cos(\theta_A - \theta_B). \tag{38}$$

Then ℓ_τ is a transport coherence length. Such a quantity requires a finite phase-memory time T_ϕ:

$$\ell_\tau = v_{\text{tr}}\, T_\phi. \tag{39}$$

At first pass, $v_{\text{tr}} = \|\mathbf{v}_{\circlearrowleft}\|$, so

$$\ell_\tau = \|\mathbf{v}_{\circlearrowleft}\| T_\phi. \tag{40}$$

Using $\omega_c = \|\mathbf{v}_{\circlearrowleft}\|/r_c$, define the phase-quality factor

$$\mathcal{Q} := \omega_c T_\phi. \tag{41}$$

Then

$$\boxed{\ell_\tau = r_c\, \mathcal{Q}.} \tag{42}$$

Dimensional check:

$$[\ell_\tau] = [r_c][\mathcal{Q}] = \text{m}.$$

Interpretation

This is the most compact structural result of the paper: coherence length is core scale times quality factor. Once ω_c is fixed canonically, the entire long-range correlation problem becomes a question about how large $\mathcal{Q}$ can be.

10 Possible SST mechanisms for phase-memory loss

A finite ℓ_τ requires a specific decoherence or decorrelation mechanism. In the broader SST setting several channels are conceivable, but for definiteness the present article adopts *topological leakage* as the worked reduced mechanism. The remaining channels are noted only briefly as alternatives for future analysis.

10.1 Density-noise dephasing

If

$$\rho_f \to \rho_f + \delta\rho,$$

then local phase frequencies fluctuate. A first-order estimate is

$$\frac{\delta\omega}{\omega_c} \sim \frac{\delta\rho}{\rho_f}, \tag{43}$$

so that

$$\mathcal{Q} \sim \frac{\rho_f}{\delta\rho}, \qquad \ell_\tau \sim r_c \frac{\rho_f}{\delta\rho}. \tag{44}$$

This possibility is not pursued further in the present paper.

10.2 Velocity-noise dephasing

If

$$\|\mathbf{v}_{\circlearrowleft}\| \to \|\mathbf{v}_{\circlearrowleft}\| + \delta v,$$

then arrival times fluctuate from cycle to cycle. Then

$$\mathcal{Q} \sim \frac{\|\mathbf{v}_{\circlearrowleft}\|}{\delta v}, \qquad \ell_\tau \sim r_c \frac{\|\mathbf{v}_{\circlearrowleft}\|}{\delta v}. \tag{45}$$

This possibility is likewise not pursued further here.

10.3 Topological leakage

If the carrier is not topologically perfect, phase information may leak into neighboring branches or through reconnection-like processes. If Γ_{topo} denotes the corresponding leakage rate, then

$$T_\phi \sim \Gamma_{\mathrm{topo}}^{-1}, \qquad \mathcal{Q} \sim \frac{\omega_c}{\Gamma_{\mathrm{topo}}}, \qquad \ell_\tau \sim \frac{\|\mathbf{v}_{\circlearrowleft}\|}{\Gamma_{\mathrm{topo}}}. \tag{46}$$

This is the mechanism adopted below for the numerical order-of-magnitude estimate, because it is the most directly tied to the persistence of shared topological carriers.

10.4 Finite clock-field coherence

If the foliation/clock sector has finite correlation length ℓ_χ, then

$$\ell_\tau \sim \ell_\chi. \tag{47}$$

This would make entanglement-range a direct property of the clock field rather than only of circulation transport. This alternative is left for future work.

A broader lesson of the reduced carrier picture is that $\mathcal{Q}$ should not be read as an abstract bookkeeping symbol only. In an SST interpretation, it is more naturally understood as a measure of how long a shared circulation carrier remains topologically and dynamically coherent while staying within a resonant or quasi-resonant operating window. This interpretation is still qualitative in the present article, but it is more restrictive than a generic noise parameter and better reflects the intended role of long-lived structured carriers.

11 First numerical implications

Using

$$r_c = 1.40897017 \times 10^{-15} \text{ m},$$

we obtain

$$\ell_\tau = r_c\, \mathcal{Q} = 1.40897017 \times 10^{-15}\, \mathcal{Q} \text{ m}.$$

Hence

$$\mathcal{Q} = 10^6 \Rightarrow \ell_\tau \approx 1.41 \times 10^{-9} \text{ m},$$

$$\mathcal{Q} = 10^{12} \Rightarrow \ell_\tau \approx 1.41 \times 10^{-3} \text{ m},$$

$$\mathcal{Q} = 10^{15} \Rightarrow \ell_\tau \approx 1.41 \text{ m},$$

$$\mathcal{Q} = 10^{21} \Rightarrow \ell_\tau \approx 1.41 \times 10^{6} \text{ m}.$$

Thus macroscopic or astronomical correlation ranges demand extremely large phase-quality factors. This is not an inconsistency; it is a hard closure requirement. Any successful SST entanglement sector must explain how such large $\mathcal{Q}$ values arise, or else predict measurable finite-range deviations from ideal quantum correlations.

12 Worked coherence estimate from topological leakage

In the reduced topological-leakage closure equation (46),

$$\mathcal{Q} \sim \frac{\omega_c}{\Gamma_{\text{topo}}}, \qquad \ell_\tau \sim \frac{\|\mathbf{v}_{\circlearrowleft}\|}{\Gamma_{\text{topo}}}.$$

These relations may be inverted to estimate the leakage rate required for a target coherence length:

$$\Gamma_{\text{topo}} \sim \frac{\|\mathbf{v}_{\circlearrowleft}\|}{\ell_\tau}. \tag{48}$$

For meter-scale coherence, $\ell_\tau \sim 1$ m, one obtains

$$\Gamma_{\text{topo}} \sim \frac{1.09384563 \times 10^6 \text{ m s}^{-1}}{1 \text{ m}} \approx 1.09 \times 10^6 \text{ s}^{-1}. \tag{49}$$

Equivalently,

$$\mathcal{Q} \sim \frac{\omega_c}{\Gamma_{\text{topo}}} \approx \frac{7.76344 \times 10^{20}}{1.09 \times 10^6} \approx 7.10 \times 10^{14}. \tag{50}$$

Thus meter-scale persistence in the present toy model would require topological leakage rates no larger than order MHz. Longer coherence scales require proportionally smaller leakage rates. This does not yet establish that such values occur in SST, but it converts the phase-quality requirement into a concrete dynamical target.

13 Linear stability of the stationary branches

The stationary branches $\Delta^* = n\pi$ identified in equation (13) admit a simple linear stability analysis. Let

$$\Delta(t) = \Delta^* + \delta(t), \qquad |\delta(t)| \ll 1. \tag{51}$$

For the in-phase branch $\Delta^* = 0$, one has $\sin(\delta) \approx \delta$, so equation (12) linearizes to

$$\dot{\delta}(t) = -2\kappa_{AB}\, \delta(t - \tau). \tag{52}$$

Seeking exponential modes $\delta(t) \propto e^{\lambda t}$ yields the characteristic equation

$$\lambda = -2\kappa_{AB} e^{-\lambda\tau}. \tag{53}$$

For the scalar delay equation $\dot{\delta}(t) = -a\,\delta(t-\tau)$ with $a > 0$, the in-phase branch is stable in the weak-delay regime when

$$a\tau < \frac{\pi}{2}. \tag{54}$$

Hence the corresponding condition here is

$$2\kappa_{AB}\tau < \frac{\pi}{2}, \qquad \text{equivalently} \qquad \mathcal{O}_{AB}\frac{L}{r_c} < \frac{\pi}{4}. \tag{55}$$

For the antiphase branch $\Delta^* = \pi$, one has $\sin(\pi + \delta) \approx -\delta$, so the linearized equation becomes

$$\dot{\delta}(t) = +2\kappa_{AB}\,\delta(t-\tau), \tag{56}$$

with characteristic equation

$$\lambda = +2\kappa_{AB} e^{-\lambda\tau}. \tag{57}$$

The controlling dimensionless combination is therefore

$$2\kappa_{AB}\tau = 2\,\omega_c \mathcal{O}_{AB}\frac{L}{\|\mathbf{v}_{\circlearrowleft}\|} = 2\,\mathcal{O}_{AB}\frac{L}{r_c}. \tag{58}$$

In the present toy-model interpretation, equation (58) shows that branch stability is controlled jointly by carrier geometry and overlap. The in-phase branch $\Delta^* = 0$ is the natural attractive branch in the weak-delay regime defined by equation (55), while the antiphase branch $\Delta^* = \pi$ is correspondingly disfavored there. A complete stability diagram for the delayed system is left for future work.

It is worth noting that several macroscopic rows of Table 1 lie far beyond the weak-delay threshold $\mathcal{O}_{AB}L/r_c < \pi/4$. Those rows should therefore be read only as parameter-scale illustrations, not as examples guaranteed to lie in the simple weak-delay locking regime.

14 Representative parameter choices and numerical regimes

To make the reduced model more concrete, it is useful to express the key control parameters in dimensionless form. The delayed phase system depends on

$$\tau\omega_c = \frac{L}{r_c}, \qquad \frac{\kappa_{AB}}{\omega_c} = \mathcal{O}_{AB}, \qquad \frac{\ell_\tau}{r_c} = \mathcal{Q}.$$

Hence the model is governed by three reduced quantities:

$$\Lambda := \frac{L}{r_c}, \qquad \mathcal{O}_{AB}, \qquad \mathcal{Q}. \tag{59}$$

Table 1 lists representative scales for several illustrative carrier lengths and overlap values. These are not claimed as unique SST predictions; they are reference regimes for reduced-model exploration.

The main lesson of Table 1 is structural rather than predictive. Carrier lengths much larger than r_c correspond to extremely large reduced delays $\Lambda = L/r_c$, while macroscopic coherence requires correspondingly large $\mathcal{Q}$. At the same time, the effective phase-locking rate is highly sensitive to $\mathcal{O}_{AB}$, so weak geometric overlap can strongly suppress coupling even when ω_c is large.

Table 1: Representative parameter choices for the reduced delayed-phase toy model. The transport delay is $\tau = L/\|\mathbf{v}_{\circlearrowleft}\|$, the coupling is $\kappa_{AB} = \omega_c \mathcal{O}_{AB}$, and the coherence length is $\ell_\tau = r_c \mathcal{Q}$. For illustration, a single reference value $\mathcal{Q} = 10^{15}$ is used throughout the final column.

L	L/r_c	τ	$\mathcal{O}_{AB}$	κ_{AB}	ℓ_τ for chosen $\mathcal{Q}$
10^{-12} m	7.10×10^{2}	9.14×10^{-19} s	1	$7.76 \times 10^{20}\,\mathrm{s}^{-1}$	1.41 m
10^{-9} m	7.10×10^{5}	9.14×10^{-16} s	10^{-1}	$7.76 \times 10^{19}\,\mathrm{s}^{-1}$	1.41 m
10^{-6} m	7.10×10^{8}	9.14×10^{-13} s	10^{-3}	$7.76 \times 10^{17}\,\mathrm{s}^{-1}$	1.41 m
1 m	7.10×10^{14}	9.14×10^{-7} s	10^{-6}	$7.76 \times 10^{14}\,\mathrm{s}^{-1}$	1.41 m

15 Numerical illustration of the reduced delayed phase model

To make the reduced toy model more concrete, we include a simple numerical integration of the delayed phase system

$$\dot{\phi}_A(t) = \omega_0 + \kappa_{AB} \sin[\phi_B(t-\tau) - \phi_A(t)], \tag{60}$$

$$\dot{\phi}_B(t) = \omega_0 + \kappa_{AB} \sin[\phi_A(t-\tau) - \phi_B(t)], \tag{61}$$

together with the reduced binary observables of equations (14) and (15). The purpose of these numerics is limited: they illustrate branch selection and the emergence of a nontrivial angle-dependent reduced correlation function $C(\theta_A, \theta_B)$ within the toy model. They are not intended as a Bell/CHSH calculation.

A minimal numerical experiment is therefore to fix ω_0, τ, and κ_{AB}, prescribe a phase history on the interval $t \in [-\tau, 0]$, and integrate the system forward while monitoring $\Delta(t)$ and the induced binary observables equations (14) and (15). In the absence of a complete Bell/CHSH structure, the most direct reduced-model outputs are:

$$\Delta(t), \qquad A(\theta_A, t), \qquad B(\theta_B, t), \qquad C(\theta_A, \theta_B).$$

In the numerical example used here, the representative integration parameters are $\omega_0 = 20\ \mathrm{s}^{-1}$, $\kappa_{AB} = 8\ \mathrm{s}^{-1}$, $\tau = 0.35$ s, $t_{\mathrm{final}} = 40$ s, and a constant history $\phi_A = 0.15$, $\phi_B = 2.10$ on the interval $[-\tau, 0]$. These values are chosen only to illustrate the dynamical behavior of the reduced mathematical model in a computationally accessible regime. They are not intended to coincide with the SST-motivated scales listed in Table 1, which are many orders of magnitude more extreme.

For the phase-locked branch $\Delta^* = 0$ and the sign-threshold observables of equations (14) and (15), one expects analytically that $C(\theta_A, \theta_B)$ reduces to the standard triangular wave obtained by overlapping the sign of two cosine functions with relative angle shift $\theta_B - \theta_A$. The numerical correlation plot should therefore be read as a consistency check of the reduced observable layer rather than as an independent discovery.

To see this explicitly, take the fully locked limit $\phi_A(t) \approx \phi_B(t) \approx \phi(t)$, and assume that the long-time dynamics sample ϕ uniformly on $[0, 2\pi)$. Then

$$C(\theta_A, \theta_B) = \frac{1}{2\pi} \int_0^{2\pi} \mathrm{sign}\big[\cos(\phi - \theta_A)\big]\, \mathrm{sign}\big[\cos(\phi - \theta_B)\big]\, d\phi. \tag{62}$$

By rotational invariance this depends only on $\Delta\theta := \theta_B - \theta_A$. Over one period, the two sign functions agree on a fraction $1 - |\Delta\theta|/\pi$ of the interval and disagree on a fraction $|\Delta\theta|/\pi$, for $|\Delta\theta| \leq \pi$. Therefore

$$C(\Delta\theta) = \left(1 - \frac{|\Delta\theta|}{\pi}\right) - \frac{|\Delta\theta|}{\pi} = 1 - \frac{2|\Delta\theta|}{\pi}, \qquad |\Delta\theta| \leq \pi, \tag{63}$$

extended periodically with period 2π. This is precisely the triangular wave seen in figure 2.

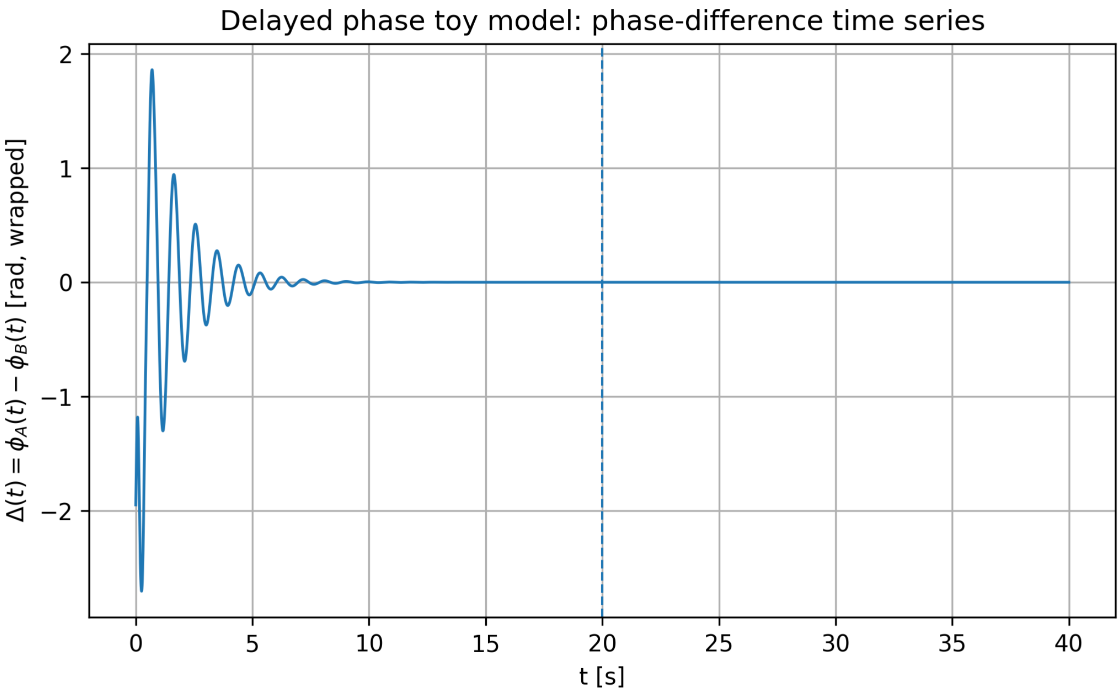

Figure 1: Time series of the wrapped phase difference $\Delta(t) = \phi_A(t) - \phi_B(t)$ for a representative delayed-phase toy-model integration. After an initial transient with damped oscillations, the system relaxes toward the $\Delta^* = 0$ branch. The dashed vertical line marks the onset of the post-transient window used for the reduced correlation analysis.

The numerical illustrations in figures 1 and 2 establish two limited but useful points. First, the analytically identified branch structure is dynamically realized under explicit delayed evolution. Second, the reduced observable layer built from thresholded phases produces a nontrivial angle-dependent correlation function. These figures therefore support the restricted claim of the paper: the delayed SST toy model admits a concrete operational readout structure, even though a full Bell/CHSH analysis remains outside the present scope.

16 Minimal effective summary

The closure program can now be stated compactly:

$$\boxed{\tau = \frac{L}{\|\mathbf{v}_{\circlearrowleft}\|}} \tag{64}$$

$$\boxed{\kappa_{AB} = \omega_c\, \mathcal{O}_{AB}} \tag{65}$$

$$\boxed{\ell_\tau = r_c\, \mathcal{Q}} \tag{66}$$

with

$$\omega_c = \frac{\|\mathbf{v}_{\circlearrowleft}\|}{r_c}. \tag{67}$$

Within the toy-model hierarchy developed above, $\mathcal{O}_{AB}$ is most naturally taken to be the energy-normalized overlap functional of equation (24), while $\mathcal{Q}$ remains a reduced parameter encoding the effective phase-memory quality of the carrier. In the thin equal-length filament limit, $\mathcal{O}_{AB}$ reduces to the simpler shared-length expression

$$\mathcal{O}_{AB} = \frac{\ell_{\text{shared}}}{L}.$$

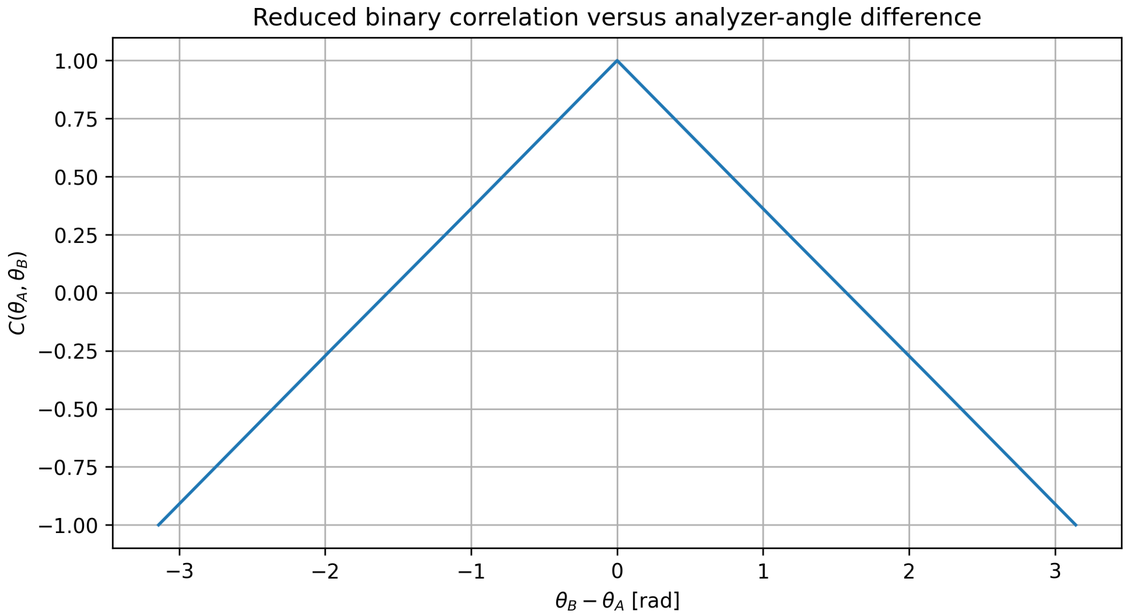

Figure 2: Reduced binary correlation $C(\theta_A, \theta_B)$ as a function of the analyzer-angle difference $\theta_B - \theta_A$, computed from the thresholded observables equations (14) and (15) after discarding the transient. In the present representative run, the toy model yields the triangular angle dependence expected analytically for a phase-locked $\Delta^* = 0$ branch together with sign-threshold observables. The numerics therefore serve as a consistency check of the reduced readout layer.

More generally, the intended SST reading is that $\mathcal{O}_{AB}$ measures carrier compatibility rather than bare contact, and that $\mathcal{Q}$ measures persistence of that compatible carrier under structured delayed transport. The long-term aim is therefore not merely to assign three free scales, but to derive them from the geometry, orientation, and dynamical quality of shared circulation structures.

17 Falsifiable consequences

The closure has three direct empirical consequences. Here again, "entanglement-like" means only the reduced operational notion introduced earlier, not a completed Bell/CHSH framework.

17.1 Nontrivial angle-dependent reduced correlations

The binary observables equations (14) and (15) define a reduced correlation function $C(\theta_A, \theta_B)$ through equation (16). Even without a Bell analysis, the toy model predicts that phase locking on a shared carrier produces nontrivial analyzer-angle dependence whenever $\kappa_{AB} \neq 0$. A null result $C(\theta_A, \theta_B) \equiv 0$ across all angles would falsify the reduced readout layer of the model.

17.2 Finite-range deviation from ideal Bell scaling

If $\mathcal{Q}$ is finite, then ℓ_τ is finite and ideal long-range correlations must weaken at sufficiently large L:

$$E(\theta_A, \theta_B) \sim e^{-L/\ell_\tau} \cos(\theta_A - \theta_B). \tag{68}$$

A null result out to arbitrarily large distances forces $\mathcal{Q} \to \infty$ or invalidates the simple closure.

17.3 Medium-sensitive dephasing

If $\mathcal{Q}$ is controlled by density or velocity fluctuations, then correlation visibility should depend weakly on environmental or medium-like perturbations. A strict null result would disfavor the noise-driven versions (44) and (45).

17.4 Topology-sensitive coherence

If $\mathcal{Q}$ is set by topological leakage, then geometrically different production channels should carry different coherence-length budgets. This predicts channel-dependent visibility loss even at fixed separation.

18 What this does and does not solve

This article solves the dimensional and structural closure problem:

$$(\tau, \kappa, \ell_\tau) \quad \Rightarrow \quad \left(\frac{L}{\|\mathbf{v}_{\circlearrowleft}\|},\ \omega_c \mathcal{O}_{AB},\ r_c \mathcal{Q} \right).$$

It does *not* yet solve the full statistical problem:

(1) deriving exact Bell/CHSH values,

(2) proving operational no-signaling,

(3) deriving Born probabilities from the underlying SST medium.

Those remain the next closure program. What the present paper does provide is a self-contained reduced model with explicit observables, an explicit geometric coupling functional, and a worked order-of-magnitude condition for maintaining long coherence.

19 Conclusion

In the present toy-model setting, the entanglement problem in SST can be reduced to a concrete triad of scales: delay, coupling, and coherence. Once the canonical identification

$$\omega_c = \frac{\|\mathbf{v}_{\circlearrowleft}\|}{r_c}$$

is accepted, the transport delay is fixed:

$$\tau = \frac{L}{\|\mathbf{v}_{\circlearrowleft}\|}.$$

Dimensional consistency then forces the phase-locking rate to be an intrinsic turnover frequency multiplied by a dimensionless shared-geometry factor:

$$\kappa_{AB} = \omega_c\, \mathcal{O}_{AB}.$$

Likewise, the correlation length is forced into the form

$$\ell_\tau = r_c\, \mathcal{Q}.$$

These relations shift the conceptual burden. In this reduced SST model, entanglement-like correlation is no longer treated purely as an abstract kinematic relation; instead, correlation strength is controlled by the degree of genuinely shared circulation structure, and correlation range by the phase-memory quality of that structure. The open problem is therefore not *whether* such parameters can be written down in a toy model, but what specific SST mechanism determines $\mathcal{O}_{AB}$ and $\mathcal{Q}$ microscopically, and whether that mechanism can recover the stronger statistical structure of quantum theory.

That is the point at which the toy model becomes sharply testable.

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